

Harvesting the Data Revolution

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Abstract

This report provides a summary of the 3rd annual conference of the National Science Foundation (NSF) Harnessing the Data Revolution (HDR) Ecosystem Conference. Held from September 9-12, 2024, at the University of Illinois Urbana-Champaign, the conference served as a key forum for the HDR community to discuss advancements and collaborations in data-intensive science. The main themes revolved around the integration of machine learning with scientific principles, addressing ethical considerations in AI, and fostering a collaborative, sustainable data science ecosystem. The event convened researchers, students, industry experts, and NSF representatives to foster collaboration, share advancements in data-intensive science and engineering, and address challenges in building a robust data ecosystem.



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Executive Summary

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1 Harnessing the Data Revolution

NSF Harnessing the Data Revolution (HDR) Big Idea is a national-scale activity founded in 2016 to enable new modes of data-driven discovery that address fundamental questions at the frontiers of science and engineering. By engaging NSF research community in the pursuit of fundamental research in data science and engineering, the ecosystem of HDR Institutes, Data Science Corps (DSC), and Transdisciplinary Research in Principles of Data Science (TRIPODS) strive to provide a cohesive, federated, national-scale approach to research data infrastructure, and the development of a 21st-century data-capable workforce.

2 Overview of 2024 Conference: *Harvesting the Data Revolution*

This conference was the 3rd annual conference of the National Science Foundation (NSF) Harnessing the Data Revolution (HDR) Ecosystem. Led by the A3D3 HDR Institute and held from September 9-12, 2024, at the University of Illinois Urbana-Champaign, the conference served as a key forum for the HDR community to discuss advancements and collaborations in data-intensive science. The main themes revolved around the integration of machine learning with scientific principles, addressing ethical considerations in AI, and fostering a collaborative, sustainable data science ecosystem. The event convened researchers, students, industry experts, and NSF representatives to foster collaboration, share advancements in data-intensive science and engineering, and address challenges in building a robust data ecosystem.

2.1 HDR Ecosystem and Prior Activities

NSF's Harnessing the Data Revolution (HDR) Big Idea is a national-scale activity founded in 2016 to enable new modes of data-driven discovery that address fundamental questions at the frontiers of science and engineering. By engaging NSF's research community in the pursuit of fundamental research in data science and engineering, the ecosystem of HDR Institutes, Data Science Corps (DSC), and Transdisciplinary Research in Principles of Data Science (TRIPODS) strive to provide a cohesive, federated, national-scale approach to research data infrastructure, and the development of a 21st-century data-capable workforce.

Members of the HDR Ecosystem have engaged in cross-cutting activities in areas such as the ML Challenges, education, workforces development, and infrastructure. The 2024 *Harvesting the Data Revolution* HDR Ecosystem conference was the 3rd annual conference series. The [first HDR Ecosystem conference](#) was held in 2022 in Alexandria, Virginia and had 77 participants. The [2023 conference](#) was led by the ID4 HDR Institute and held in Denver, Colorado with 100 participants.

2.2 Venue

The conference venue was the Illinois Conference Center which is connected to the I-HOTEL, a contemporary, AAA Four-Diamond rated hotel and event venue located on the southwest part of the campus of the University of Illinois Urbana-Champaign. Opened in 2009, the I-Hotel and Illinois Conference Center facilities feature 125 hotel rooms, a restaurant and bar, and over 70,000 square feet of conference space fully equipped with modern technology. It is designed with a focus on blending art, nature, and wellness, offering a serene and sophisticated environment for travelers, business professionals, and event hosts. The University of Illinois Urbana-Champaign is a public land-grant research university in the Champaign–Urbana metropolitan area, Illinois, United States. It is the flagship institution of the University of Illinois system and was established in 1867. With over 56,000 students, the University of Illinois is one of the largest public universities by enrollment in the United States. The university contains 16 schools and colleges and offers more than 150 undergraduate and over 100 graduate programs of study. The university holds 651 buildings on 6,370 acres and its annual operating budget in 2023 was over \$7.7 billion. The campus library system possesses the fourth-largest university library in the United States by holdings. The university also hosts the National Center for Supercomputing Applications (NCSA), supported by the National Science Foundation, that provides leading-edge computing, data storage, and visualization resources. The University of Illinois Urbana-Champaign also operates a Research Park home to innovation centers for over 120 start-up companies and multinational corporations.

The conference was mostly for in-person attendance but [Zoom](#) was also available to remote participants. The importance of this hybrid conference format for reducing the carbon footprint of our research activities, while also enabling greater diversity and inclusion for participation, is described in [1].

2.3 Organizers

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2.4 Participants

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2.5 Objectives

Its primary objectives were to:

- **Strengthen the ecosystem** by fostering relationships and sharing progress among HDR entities.
- **Address shared challenges** and identify opportunities for synergistic activities in data-intensive research.
- **Assess the broader landscape** of data science and AI to position the HDR ecosystem as a central component of national data-intensive research.
- **Formally launch** the HDR Machine Learning Challenge.

3 Activities

Fig. 1 shows a high-level overview conference schedule and timeline of activities.

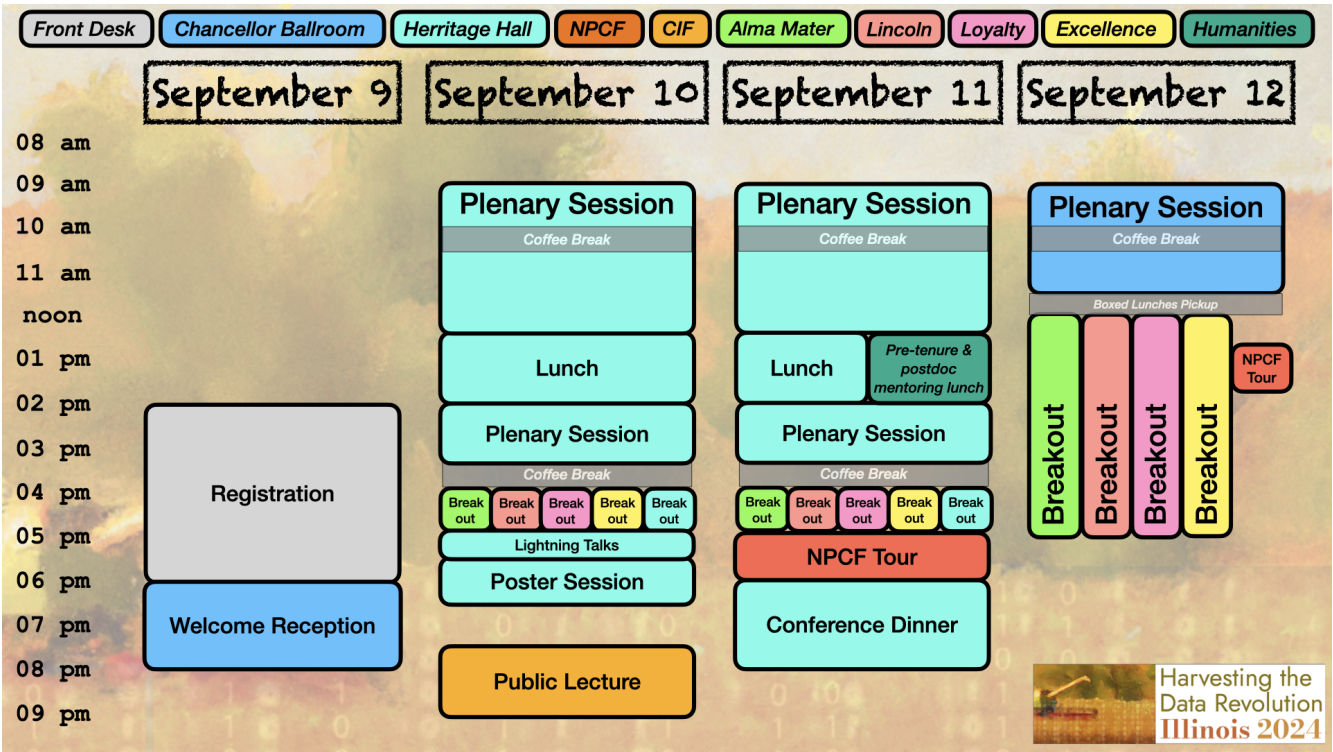


Figure 1. Overview of schedule and timeline of the 2024 HDR conference activities

3.1 Welcoming

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3.2 Plenary Sessions

3.2.1 Research

The HDR covers a vast scope of scientific problems. Here we review the different domains. Despite the very large differences present, there is a common underlying theme to how we approach research and deal with data. We highlight these underlying challenges, and approaches in the following section.

3.2.2 ML Challenges

A uniting component of our approach to analyzing and interpreting our data is the use of Machine Learning to process the data and build core scientific results. To unite the HDR institutes, we are creating ML challenges with common machine-learning themes that span the many scientific domains covered by the HDR program. In addition to the creation of the challenges, the use of FAIR and CARE data principles is fundamental to their effective deployment and possibly use in future ML benchmarks.

3.2.3 Community Engagement

To enable larger adaptation of robust data principles and to highlight the growing need for ML integration, we aim to engage the larger scientific community with our efforts. Beyond the scientific community, we aim to connect with industry members to build a partnership that can help augment our scientific results and improve knowledge transfer to many commercial domains that benefit from this research.

3.2.4 AI & Data Infrastructure

Computing infrastructure is a core component of analyzing and storing data. Adapting to the rapidly evolving data processing strategies that have resulted from advancements in AI requires many challenges. In this section, we talk about the integration of heterogeneous computing resources. Strategies for data distribution and processing, and connect these efforts with the greater national computing efforts aimed at building next-generation computing infrastructure for scientific discovery.

3.2.5 Workforce Development, Education & Outreach

To adapt to the influx of data, new skills are needed to process and curate data. Many of these skills are not present in our current education, curricula but are becoming increasingly important for many different scientific domains. In this section, we talk about strategies to develop these skills through new education initiatives, outreach, workforce development, and tutorials.

3.3 Keynote Talk and Public Lecture

The conference featured two notable speakers who set the intellectual tone for the event.

3.3.1 Vipin Kumar: Knowledge-Guided Machine Learning

Vipin Kumar, a professor at the University of Minnesota, delivered the keynote address on **Knowledge-Guided Machine Learning (KGML)**. He introduced KGML as a new paradigm for scientific discovery, where scientific knowledge and first-principle models are deeply embedded within machine learning algorithms. Kumar argued that this integration is crucial for addressing complex global environmental challenges, such as climate change and biodiversity loss, as it leads to more physically consistent and interpretable models than traditional data-driven approaches alone.

3.3.2 Suresh Venkatasubramanian: AI Policy

Suresh Venkatasubramanian from Brown University gave a public lecture on the critical topic of **AI policy**. His talk explored the challenges of making AI systems safe, effective, and trustworthy. This lecture underscored the importance of responsible AI development and the need for thoughtful policy to govern the application of AI, especially as it becomes more integrated into scientific research and society.

3.4 Expert Panels

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3.5 Unconference Open Space

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3.6 Pre-tenure Faculty & Postdoc Mentoring Luncheon

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3.7 Breakout Sessions

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3.8 Lightning Talks & Posters

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3.9 National Petascale Computing Facility Tours

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4 Highlights, Findings and Outcomes

Key outcomes included:

- Presentation and discussion of current HDR research, infrastructure, and education/workforce efforts.
- Launch (or formalization) of the HDR Machine Learning (Anomaly Detection) Challenge.
- Panels on public policy, trust and fairness in AI, infrastructure, community engagement, and workforce development.
- A public lecture and keynote emphasizing both the societal implications of data and AI, and technical frameworks for accelerating scientific discovery.
- Practical coordination: lodging, venue planning, and the compilation of shared objectives toward sustaining and growing the HDR ecosystem.

4.1 Research

TBD

4.2 ML Challenges

TBD

4.3 Community Engagement

TBD

4.4 AI & Data Infrastructure

TBD

4.5 Workforce Development, Education & Outreach

TBD

5 Looking Forward

TBD

6 Acknowledgments

We thank the conference attendees for their active participation toward achieving the conference goals, discussing key challenge and opportunities for learning from scientific data, and developing plans going forward. This conference was supported by the National Science Foundation under TI-2303700 and Cooperative Agreement OAC-2117997.

7 Disclaimer

Any opinions, findings, conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation.

Glossary

HDR Harnessing the Data Revolution <https://www.nsfhdr.org>. 2

Zoom A video conferencing platform. <https://zoom.us>. 2

References

1. M. S. Neubauer et al., “Learning from the Pandemic: the Future of Meetings in HEP and Beyond”, in *IRIS-HEP Blueprint Workshop*. 6, 2021. [arXiv:2106.15783](https://arxiv.org/abs/2106.15783).